

Arithmetic Local–Global Dichotomies for ProveIt’s Three-Variable Keller Map

Split integral Hasse failures, a dense parameter family,
and exact p -adic fiber laws

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Abstract

We study the arithmetic image of the explicit polynomial map with constant Jacobian determinant -2 developed in ProveIt’s Jacobian-conjecture project. The complex geometry and finite-field fiber laws are prior results ([write] of Gao and of the `royvanrijn/jacobian-research` notes respectively; neither is part of ProveIt); our focus is the distinction between rational, integral, and locally integral inverse images. We construct an explicit family of targets having exactly three rational preimages, none integral, but at least one integral preimage over $\mathbb{Z}_p$ for every prime p . More generally, a two-parameter split-root family admits an exact greatest-common-divisor criterion for local solubility. Its locally soluble parameters have density $27/(4\pi^2)$ and its targets are Zariski dense in the plane with third coordinate 2. The failure persists over every prescribed ring of S -integers and under integral stable equivalence.

At the exceptional prime 2, we prove that integral fiber cardinality is determined modulo 8, and has the Haar probability law $(21, 7, 3, 1)/32$ for cardinalities 0, 1, 2, 3. Consequently the 2-adic image has measure $11/32$. A general inverse-ball lemma turns a small finite certificate into a theorem at every precision. In contrast, the rational image satisfies a Hasse principle over every number field, by a cubic Galois argument. We also give a quantitative density-zero bound for rationally soluble integral targets and a precise formalization roadmap. All new claims have explicit proofs; computational identities and finite tables are supplied as replayable certificates, not represented as Lean proofs.

Status. The split integral family, its parameter density, and the dyadic refinement are candidate new contributions relative to the sources reviewed here. Global priority has not been established. This is not a new disproof of the Jacobian conjecture, nor a solution of the plane Jacobian problem.

[write] Batch 36, manuscript 03, placed in 1a1396d4d: a report of ProveIt’s research-report collection that continues the formal project `Algebra/JacobianConjecture`. AI-assisted, unrefereed, not formalized; labels carry the prefix `alg:`. See Section 1.3 and Appendix C.

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1 The research question and its exact scope

1.1 From a formal polynomial witness to an arithmetic problem

ProveIt contains independent Lean and Rocq/Coq developments for a three-variable polynomial map announced by Levent Alpöge, including its constant Jacobian determinant, explicit collisions, symmetry, and stable changes of variables [1, 2]. We use the following compact presentation, over $\mathbb{Z}$:

$$\begin{aligned} u &= 1 + xy, & h &= u^2z + y^2(1 + 3u), \\ F(x, y, z) &= (P, Q, R) = (uh, y + 3xh, x(5 - 3u - x^2z)). \end{aligned} \tag{1.1}$$

Thus

$$\begin{aligned} P &= (1 + xy)^3z + y^2(1 + xy)(4 + 3xy), \\ Q &= y + 3x(1 + xy)^2z + 3xy^2(4 + 3xy), \\ R &= 2x - 3x^2y - x^3z. \end{aligned}$$

The formal identity $\det JF = -2$ makes F an *étale* morphism over $\mathbb{Z}[1/2]$: there the Jacobian determinant is a unit. Over $\mathbb{Z}$ it is not étale at 2. This distinction will matter in the local counting theorem, but it will *not* explain away the global integral obstruction.

A collision establishes noninjectivity. It does not answer which integral targets occur as values, how much congruences detect, or whether rational and integral local–global principles agree. We therefore pose the concrete question

$$\mathbf{b} \in \mathbb{Z}^3, \quad \mathbf{b} \in F(\mathbb{Z}_p^3) \text{ for every prime } p \stackrel{?}{\implies} \mathbf{b} \in F(\mathbb{Z}^3). \tag{1.2}$$

We answer it negatively by explicit complete fibers. We then classify a large family of such failures and determine the local image at the unique bad prime.

1.2 What is inherited, and what is contributed here

The choice of this problem followed a review of the actual repository statements rather than an assumption that every unexplained phenomenon was open. In particular, Gao’s Theorems 3.3–3.4 already give the generic geometric degree and complete complex fiber stratification of this map [3]. The archived note *Exact finite-field value distribution* in `royvanrijn/jacobian-research` already gives the cubic inverse model and exact finite-field histograms, including characteristic 3 [4]. We reproduce short proofs because the arithmetic arguments depend on their precise boundary conditions, not because these are being claimed as new theorems.

There is also prior work on a *different* integral Jacobian-one map with a locally integral fiber of geometric degree five having *no rational point* [5]. Our examples instead have three rational points, all of which fail global integrality. Consequently they neither supersede that stronger rational obstruction nor conflict with its minimum-degree discussion. The two constructions isolate different failures of local–global reasoning.

Statement	Role in this article	Evidence boundary
Map F , determinant, collisions	Repository input; rederived	Prior formal developments; exact identity here
Complex fiber sizes 3, 1, 0	Prior geometry	Gao; self-contained root proof here
Odd finite-field histograms	Prior arithmetic input	Archived note; proof and replay here
Split integral Hasse family and density	Candidate new refinement	Explicit formulas and elementary proofs
S -integer and stable-equivalence persistence	Consequences of the family	Exact algebraic arguments
Dyadic law (21, 7, 3, 1)/32	Candidate new refinement	Inverse-ball theorem plus finite certificate
Rational Hasse principle; thin image bound	Applications, not priority claims	Cubic algebra, Chebotarev, elementary sieve

Our source review is dated September 24, 2026, with ProveIt pinned to e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9 and the second research repository pinned to caf5685db1099cb90e608556d4b8d825734f29cb. The review did not locate the same split-plane or dyadic statements in the inspected material. That is a bounded literature finding, not a proof of worldwide novelty. Nor do we claim that the question in (1.2) was a previously published named open problem. It is a natural unresolved arithmetic question arising from the repo, and the proofs below resolve it for this map.

1.3 Attribution and the formal project

[write] This subsection was added when the manuscript was written into ProveIt. It makes the attribution of the inherited material precise.

- *The map* was announced by Alpöge. ProveIt’s Jacobian project formalizes it, in Lean 4 as `counterexample` (`Algebra/JacobianConjecture/Lean/JacobianConjecture/Counterexample.lean`, lines 95–101, namespace `LeanProofs.JacobianCounterexample`) and in Rocq/Coq as `counterexample` (`Algebra/JacobianConjecture/Coq/Counterexample.v`, line 110), with exactly the coordinates P, Q, R displayed below.
- *The determinant and the collisions* are kernel-checked there: `jacobianDet_counterexample` ($\det JF = -2$ as a polynomial identity over every commutative ring), the integral collision `collision` with `collision0 = (-1, 1, 5)`, `collision1 = (0, -2, -16)` and `collisionValue = (0, -2, 0)`, the rational triple collision `counterexample_triple_collision` over $(-\frac{1}{4}, 0, 0)$, the mirror symmetry `counterexample_equivariant`, the torus action `counterexample_scaling` $F(x/s, sy, s^2z) = (s^2P, sQ, R/s)$, the one-parameter collision family `collisionFamily`, and the refutation `jacobianConjectureInDimensionThree_false`; the Coq development proves the corresponding statements (`jacobian_det_is_minus_two`, `counterexample_scaling`, `jacobian_conjecture_dimension_three_is_false`).
- *The complex fiber stratification* (fibers of size 3, 1, 0; [corollary 2.4](#)) is Gao’s [3, Theorems 3.3–3.4]. It is *not* in ProveIt. At the write it was checked on arXiv that Gao’s Theorem 3.3 concerns this very map (component degrees 7, 6, 4, determinant -2 , three points in the generic fiber) and that his polynomial $c_3 = 27v_1^2v_3^2 - 18v_1v_2v_3 + 16v_1 + v_3^3v_3 - v_2^2$ is $-\Delta/4$ in the notation of (2.9).

- *The cubic inverse model and the finite-field histograms* ([theorem 7.1](#) and the count [\(7.4\)](#)) are from the archived note [\[4\]](#) of [royvanrijn/jacobian-research](#), which already uses the variable $t = y + 1/x$ and the cubic [\(2.4\)](#). They are *not* in ProveIt. At the write it was checked that the pinned note states the formulas [\(7.1\)](#) and [\(7.2\)](#).
- *Nothing arithmetic about the fibers* is in ProveIt: the project has no material on fiber stratification, finite fields, p -adic or dyadic fibers, or Hasse principles, at the pin and at the placement commit. The theorems of [sections 3 to 6](#) and [8 to 10](#) are this manuscript’s (with the priority caveats stated above), and *none of them is formalized*. The stable representatives used in [proposition 5.3](#) are the project’s: its degree-six one is kernel-checked (`simplerCounterexample`), the lower-degree ones have SymPy certificates only.

1.4 Notation and the headline results

We write $\mathbf{b} = (A, B, C)$ for a target. The letter p denotes a rational prime, while q is a finite field’s cardinality. Haar measure on $\mathbb{Z}_p^d$ is normalized to give the whole space measure one. For a finite set S of primes, $\mathbb{Z}[S^{-1}]$ denotes the rationals whose denominators use only primes in S .

The main explicit example is especially small:

$$F^{-1}(0, -4, 2)(\overline{\mathbb{Q}}) = \left\{ \left(-\frac{1}{2}, 2, 36 \right), \left(\frac{1}{6}, -4, -288 \right), \left(\frac{1}{3}, -4, 0 \right) \right\}. \quad (1.3)$$

Every prime sees an integral point in this fiber, but no point is integral over $\mathbb{Z}$. The complete proof, including exhaustiveness, is in [section 3](#). The larger split-plane family and density theorem follow in [section 4](#). The dyadic law is proved in [section 9](#):

$$\Pr_{\mathbf{b} \in \mathbb{Z}_2^3} (\#F^{-1}(\mathbf{b})(\mathbb{Z}_2) = j) = \frac{1}{32} (21, 7, 3, 1)_j, \quad j = 0, 1, 2, 3.$$

Both results depend on distinguishing actual source points from roots of an inverse polynomial which have escaped to infinity.

2 A complete simple-root model

2.1 An elementary coordinate change

The following calculation supplies all the inverse formulas needed later. It is valid over any field of characteristic different from 2. On the source chart $x \neq 0$, introduce

$$w = \frac{1}{x}, \quad t = y + \frac{1}{x}, \quad C = R(x, y, z). \quad (2.1)$$

Then $u = xt$, and solving the definition of R gives

$$x = \frac{1}{w}, \quad y = t - w, \quad z = 5w^2 - 3tw - Cw^3. \quad (2.2)$$

Direct substitution into [\(1.1\)](#) now yields the much smaller formulas

$$A = wt + t^2 - Ct^3, \quad B = 2w + 4t - 3Ct^2, \quad R = C. \quad (2.3)$$

For example, substitution first gives $u = t/w$ and $h = w^2 + tw - Ct^2w$, from which both first coordinates follow immediately. Eliminating w gives

$$g_{A,B,C}(T) = CT^3 - 2T^2 + BT - 2A, \quad g_{A,B,C}(t) = 0, \quad (2.4)$$

and the derivative is

$$d(t) := g'_{A,B,C}(t) = 3Ct^2 - 4t + B = 2w = \frac{2}{x}. \quad (2.5)$$

In particular, every finite source point on this chart determines a *simple* root. A repeated root cannot be reconstructed by dividing by its derivative.

Proposition 2.1 (A short determinant proof). *As an identity in $\mathbb{Z}[x, y, z]$, one has $\det JF = -2$.*

Proof. In the localization where x is invertible, the change $(x, y, z) \mapsto (w, t, C)$ has determinant $(-x^{-2}) \cdot 1 \cdot (-x^3) = x = 1/w$. From (2.3), the determinant of $(w, t, C) \mapsto (A, B, C)$ is

$$\det \begin{pmatrix} t & w + 2t - 3Ct^2 & -t^3 \\ 2 & 4 - 6Ct & -3t^2 \\ 0 & 0 & 1 \end{pmatrix} = -2w.$$

The chain rule gives -2 in the localization. Since $\mathbb{Z}[x, y, z]$ injects into that localization, it gives the same polynomial identity globally. This is a derivation of the identity, independent of the accompanying symbolic expansion check. $\square$

Theorem 2.2 (Complete field-valued inverse). *Let k be a field with $\text{char } k \neq 2$, and $(A, B, C) \in k^3$. Every simple root $t \in k$ of $g_{A,B,C}$ determines exactly one preimage with $x \neq 0$, namely*

$$w = \frac{g'_{A,B,C}(t)}{2}, \quad (x, y, z) = \left(\frac{1}{w}, t - w, 5w^2 - 3tw - Cw^3 \right). \quad (2.6)$$

These are all the preimages with $x \neq 0$. There is a preimage with $x = 0$ exactly when $C = 0$, and then it is uniquely

$$(0, B, A - 4B^2). \quad (2.7)$$

Equivalently, the complete fiber is in natural bijection with the simple k -rational projective roots of

$$G_{A,B,C}(T, S) = CT^3 - 2T^2S + BTS^2 - 2AS^3. \quad (2.8)$$

Proof. The forward construction and simplicity follow from (2.1)–(2.5). Conversely, set $w = g'(t)/2 \neq 0$. The second expression in (2.3) is B . The first expression minus A equals $g(t)/2$ after this substitution, so it vanishes. This proves that (2.6) is a two-sided inverse on the chart. If $x = 0$, (1.1) reduces to $F(0, y, z) = (z + 4y^2, y, 0)$, proving (2.7). Finally, the projective point $[1 : 0]$ is a root of G exactly when $C = 0$. On the chart $T = 1$, its derivative with respect to S is -2 , so it is always simple. All other projective roots are the affine roots already classified. $\square$

Remark 2.3 (Why a root count can be wrong). A cubic may have three roots counted with multiplicity while the actual fiber has only one point, or none. Also, setting $C = 0$ lowers the degree of the affine inverse polynomial but *adds* the source point (2.7). Both issues must be handled before making any arithmetic inference. In particular, a resultant equation alone is not an exhaustive inverse description.

2.2 The known geometric stratification

The discriminant of the binary cubic, or of g with its cubic coefficient retained, is

$$\Delta = 4B^2 - 4CB^3 - 64A + 72CAB - 108C^2A^2. \quad (2.9)$$

Over an algebraically closed field of characteristic different from 2 and 3, [theorem 2.2](#) immediately gives the following result.

Corollary 2.4 (Previously known complex fiber law). *There are three preimages if $\Delta \neq 0$, one if $\Delta = 0$ but the binary cubic is not a cube, and none on the triple-root locus*

$$C \neq 0, \quad 3BC = 4, \quad 27AC^2 = 4. \quad (2.10)$$

There is no fiber of cardinality two over an algebraically closed field.

Proof. A separable degree-three divisor on $\mathbb{P}^1$ has three simple points. A degree-three divisor with a repeated point, but not supported at a single point, has one double point and one simple point. A cube has no simple point. Comparing $C(T - t_0)^3$ with g gives $t_0 = 2/(3C)$ and then (2.10). A triple root at infinity is impossible because the coefficient of T^2S is -2 . $\square$

This recovers Gao’s stratification [3, Theorem 3.4]; indeed his polynomial c_3 is $-\Delta/4$. We use a different inverse coordinate because the derivative identity (2.5) is especially convenient for denominators. The geometric law does not imply that every *integral* fiber has size 0, 1, or 3; integrality can discard just one of three rational points.

2.3 The projective chart at infinity

For completeness, the projective-root picture can be made into an explicit scheme description over $\mathbb{Z}[1/2]$. Let Y be the incidence subscheme $G = 0$ in $\mathbb{A}^3 \times \mathbb{P}^1$, and let Y^{simp} be the open set where the corresponding root is simple. The map from the source is

$$(x, y, z) \mapsto (F(x, y, z), [1 + xy : x]).$$

The projective pair is globally defined because $(1 + xy) - yx = 1$. On $S \neq 0$, [theorem 2.2](#) gives its inverse. On $T \neq 0$, put $s = S/T$; the incidence equation becomes

$$C = 2s - Bs^2 + 2As^3.$$

Set

$$k = 1 - Bs + 3As^2, \quad Y_0 = B - 3As.$$

Simplicity is precisely $k \neq 0$, because the derivative of $C - 2s + Bs^2 - 2As^3$ is $-2k$. An inverse chart is

$$x = \frac{s}{k}, \quad y = Y_0, \quad z = Ak^3 - k(k + 3)Y_0^2. \quad (2.11)$$

Indeed $1 + xy = 1/k$, $h = Ak$, and the three coordinates of F become $A, B, 2s - Bs^2 + 2As^3$. These identities prove that the two chart inverses glue and that the source is isomorphic to Y^{simp} . This is a coordinate interpretation, not a separate novelty claim. No scheme-theoretic machinery from this paragraph is needed in the integral arguments below.

3 An explicit infinite integral Hasse failure

3.1 The family and all its points

Theorem 3.1 (A split three-point obstruction). *For every integer $n \geq 2$, let*

$$\mathbf{b}_n = (0, -2n(n-1), 2). \quad (3.1)$$

Then

$$\mathbf{b}_n \in F(\mathbb{Z}_p^3) \text{ for every prime } p, \quad \mathbf{b}_n \in F(\mathbb{Q}^3) \subset F(\mathbb{R}^3), \quad \mathbf{b}_n \notin F(\mathbb{Z}^3).$$

The complete geometric fiber consists of three rational points. More precisely, with

i	t_i	D_i
0	0	$-n(n-1)$
1	n	$n(2n-1)$
2	$1-n$	$(n-1)(2n-1)$

(3.2)

those points are

$$v_i(n) = \left(D_i^{-1}, t_i - D_i, 5D_i^2 - 3t_i D_i - 2D_i^3 \right). \quad (3.3)$$

Proof. The inverse cubic factors exactly as

$$g_{\mathbf{b}_n}(T) = 2T^3 - 2T^2 - 2n(n-1)T = 2T(T-n)(T+n-1).$$

For $n \geq 2$, its roots $0, n, 1-n$ are distinct. Half its derivative at these roots is respectively the three values in (3.2). Since the third target coordinate is nonzero, there is no $x = 0$ point. The complete inverse theorem therefore gives exactly (3.3), even over $\overline{\mathbb{Q}}$.

All second and third coordinates of these points are integers. Their first coordinates are $1/D_i$, with $|D_i| > 1$, so none is an integer point. This proves global insolubility without a bounded search.

The three integers $n, n-1, 2n-1$ are pairwise coprime: subtracting adjacent integers proves the first assertion, and $2n - (2n-1) = 1$, $(2n-1) - 2(n-1) = 1$ prove the other two. A prime divides at most one of them. If it divides n , select v_2 ; if it divides $n-1$, select v_1 ; if it divides $2n-1$, select v_0 . The selected denominator is a p -adic unit. If it divides none, all three points are p -adically integral. Thus every prime has an integral preimage. The displayed points themselves are rational and real. $\square$

Example 3.2 (The smallest displayed parameter). At $n = 2$, the points are exactly those in (1.3). At the prime 2, use $(1/3, -4, 0)$. At the prime 3, use $(-1/2, 2, 36)$. At every other prime all three points are integral. A single rational point integral at every prime would be a $\mathbb{Z}$ -point, but here the locally selected point changes with the prime. The quantifiers

$$\forall p \exists v_p \quad \text{and} \quad \exists v \forall p$$

are genuinely different, despite the fiber having only three points.

3.2 A denominator criterion for every split finite fiber

The preceding mechanism is elementary and reusable.

Lemma 3.3 (Denominator covering). *Let $f \in \mathbb{Z}[X_1, \dots, X_d]^d$ and $\mathbf{b} \in \mathbb{Z}^d$. Suppose its complete geometric fiber is a finite nonempty set of rational points $v_1, \dots, v_r$. Let d_i be the least positive common denominator of the coordinates of v_i . Then*

$$\begin{aligned} \mathbf{b} \in f(\mathbb{Z}^d) &\iff d_i = 1 \text{ for some } i, \\ \mathbf{b} \in f(\mathbb{Z}_p^d) &\iff p \nmid d_i \text{ for some } i, \\ \mathbf{b} \in f(\mathbb{Z}_p^d) \text{ for every } p &\iff \gcd(d_1, \dots, d_r) = 1. \end{aligned}$$

Proof. A rational vector is integral over $\mathbb{Z}_p$ exactly when its reduced coordinate denominators are all prime to p . Because every geometric point is rational, extension to $\mathbb{Q}_p$ creates no additional points in the fiber. Hence local membership is exactly the stated disjunction over the existing finite set. A prime fails every disjunct precisely when it divides their greatest common divisor. The global assertion is the corresponding denominator-one condition. $\square$

The split hypothesis matters: a nonsplit factor can acquire new points over some completions. We will not silently apply the lemma to a cubic with an irreducible quadratic factor.

3.3 An actual solution of every congruence

Corollary 3.4. *For every $n \geq 2$ and every positive integer m , there is an integer vector a with $F(a) \equiv \mathbf{b}_n \pmod{m}$, although $F(a) = \mathbf{b}_n$ has no integer solution.*

Proof. For each prime power $p^e \mid m$, select a point $v_i(n)$ with $p \nmid D_i$. Reduce its integer coordinates normally and its first coordinate by $D_i^{-1} \pmod{p^e}$. This gives a source modulo p^e . Combine these source coordinates using the Chinese remainder theorem. Since F has integer coefficients, its value has the required reduction at every prime power. $\square$

The accompanying function `congruence_preimage(n,m)` performs exactly this construction. For example,

$$F(20587, 20252, 13536) \equiv (0, -4, 2) \pmod{21600}.$$

This is an exact certificate, not a numerical approximation to a nonexistent integer solution.

4 A dense two-parameter family and its exact density

4.1 Complete classification on the integral split-root locus

The one-parameter family is the specialization $(a, b, c) = (0, n, 1 - n)$ of a much larger construction.

Theorem 4.1 (Split-plane classification). *Let $a, b, c \in \mathbb{Z}$ be pairwise distinct with $a + b + c = 1$, and define*

$$\Phi(a, b) = (abc, 2(ab + ac + bc), 2), \quad c = 1 - a - b. \quad (4.1)$$

For $r \in \{a, b, c\}$ put

$$D_r = \prod_{s \in \{a, b, c\}, s \neq r} (r - s).$$

Then the complete geometric fiber of F over $\Phi(a, b)$ consists of

$$\left(D_r^{-1}, r - D_r, 5D_r^2 - 3rD_r - 2D_r^3\right), \quad r = a, b, c. \quad (4.2)$$

It has no integral point. Writing

$$g = \gcd(a - b, a - c) = \gcd(a - b, 3a - 1), \quad (4.3)$$

we have

$$\Phi(a, b) \in F(\mathbb{Z}_p^3) \text{ for every prime } p \iff g = 1. \quad (4.4)$$

More precisely, a prime has no integral preimage exactly when it divides g .

Proof. The coefficient identities for three roots of sum 1 give

$$g_{\Phi(a,b)}(T) = 2(T - a)(T - b)(T - c).$$

Thus [theorem 2.2](#) gives (4.2), with no boundary point. If $|D_r| = 1$, both other roots must be integers at distance one from r . Because they are distinct, they must be $r - 1$ and $r + 1$. Their sum with r would be $3r = 1$, an impossibility for integral r . Hence all three first coordinates are nonintegral.

For the exact gcd, put $u = a - b$ and $v = a - c$. Up to signs the three D 's are

$$uv, \quad u(v - u), \quad v(v - u).$$

After writing $u = gu_0$, $v = gv_0$, the integers $u_0, v_0, v_0 - u_0$ are pairwise coprime. No prime can divide all three of their pairwise products. Consequently

$$\gcd(|D_a|, |D_b|, |D_c|) = g^2. \quad (4.5)$$

All other coordinates in (4.2) are integral. Apply [lemma 3.3](#). Finally, $(a - b) + (a - c) = 3a - 1$, proving the second expression for g . $\square$

Remark 4.2 (Exact local multiplicities in this family). If $p \nmid g$ and exactly two roots agree modulo p , only the point corresponding to the third root is integral over $\mathbb{Z}_p$. If all three root residues are distinct, all three points are integral. If $p \mid g$, no point is integral. In particular, for every locally soluble target of this family, the integral fiber has exactly one point at $p = 2$ and at $p = 3$. At 2 three distinct residues are impossible. At 3 three distinct residues would sum to 0, whereas the root sum is 1.

4.2 A positive density theorem in the parameter plane

Density here refers to ordered integer parameters (a, b) , not to all integer targets in $\mathbb{A}^3$. The distinction is essential.

Theorem 4.3 (Parameter density). *Let $M(H)$ count pairs $(a, b) \in [-H, H]^2 \cap \mathbb{Z}^2$ such that the three integers $a, b, 1 - a - b$ are distinct and $\gcd(a - b, 3a - 1) = 1$. As $H \rightarrow \infty$ through positive integers,*

$$M(H) = \frac{27}{4\pi^2}(2H + 1)^2 + O(H \log H). \quad (4.6)$$

Every counted pair gives a locally integral but globally nonintegral fiber of the fixed map F .

Proof. First ignore coincidences among the roots, and let $L = 2H + 1$. Since $3a - 1 \neq 0$, the gcd in question is positive and at most $3H + 1$. The Möbius identity for coprimality gives the count

$$\sum_{d \leq 3H+1} \mu(d) \# \{(a, b) \in [-H, H]^2 : d \mid a - b, d \mid 3a - 1\}.$$

The inner set is empty when $3 \mid d$. Otherwise 3 is invertible modulo d , and there is exactly one residue pair:

$$a \equiv b \equiv 3^{-1} \pmod{d}.$$

Its count is $L^2/d^2 + O(L/d + 1)$, uniformly in d . Summing the errors gives $O(H \log H)$, while extending the absolutely convergent main sum to infinity costs $O(H)$. Its value is

$$\sum_{3 \nmid d} \frac{\mu(d)}{d^2} = \prod_{p \neq 3} \left(1 - \frac{1}{p^2}\right) = \frac{1}{\zeta(2)(1 - 3^{-2})} = \frac{27}{4\pi^2}.$$

The three root-coincidence equations $a = b$, $2a + b = 1$, and $a + 2b = 1$ account for only $O(H)$ pairs. Removing them does not affect the displayed asymptotic. The last assertion is [theorem 4.1](#). $\square$

The approximate value of the density is 0.683918. Exact enumeration, used only as a check on the proof, gives the following data.

H	$M(H)$	$M(H)/(2H + 1)^2$
10	310	0.702948
25	1764	0.678201
50	7034	0.689540
100	27706	0.685775
250	171878	0.684770

Corollary 4.4 (Zariski density in a fixed target plane). *The set*

$$\{\Phi(a, b) : a, b, 1 - a - b \text{ distinct, } \gcd(a - b, 3a - 1) = 1\}$$

is Zariski dense in the plane $C = 2$. Every target in it is an integral Hasse failure for F in the sense of (1.2).

Proof. A nonzero polynomial in two variables has only $O(H)$ zeros in $[-H, H]^2 \cap \mathbb{Z}^2$: view it as a polynomial in one variable and separate the finitely many values where all coefficient polynomials vanish. By [theorem 4.3](#), the allowed parameter pairs have positive two-dimensional density, so are Zariski dense in $\mathbb{A}^2$.

The map Φ is dominant onto the target plane. Indeed, for any $(A, B) \in \overline{\mathbb{Q}}^2$, the monic cubic

$$T^3 - T^2 + \frac{B}{2}T - A$$

has three roots over $\overline{\mathbb{Q}}$, counted with multiplicity, whose sum is 1; those roots give a parameter pair mapping to $(A, B, 2)$. A polynomial relation vanishing on the allowed target set would therefore pull back to a relation vanishing on a dense parameter set, hence would vanish identically on the target plane. $\square$

Remark 4.5 (What the density does not say). The map from ordered distinct root triples to a fixed target has six permutations. Moreover, a parameter box has coefficient heights $|A| = O(H^3)$ and $|B| = O(H^2)$, not a uniform target box. Thus (4.6) is not a density formula in target space. It does prove that the integral obstruction is much more than an isolated collision or a single exceptional arithmetic progression.

5 S -integers, stable equivalence, and congruence closure

5.1 Inverting finitely many primes does not remove the obstruction

Theorem 5.1 (Persistence over every prescribed S). *For every finite set S of rational primes, infinitely many $n \geq 2$ satisfy*

$$\mathbf{b}_n \in F(\mathbb{Z}_p^3) \text{ for every } p, \quad \mathbf{b}_n \notin F(\mathbb{Z}[S^{-1}]^3).$$

In particular, this failure persists for the étale map over $\mathbb{Z}[1/2]$.

Proof. Choose distinct primes ℓ_1, ℓ_2 outside S . By the Chinese remainder theorem, there are infinitely many $n \geq 2$ satisfying $n \equiv 0 \pmod{\ell_1}$ and $n \equiv 1 \pmod{\ell_2}$. In (3.2), the denominators D_0, D_1 are divisible by ℓ_1 , and D_2 is divisible by ℓ_2 . Their numerators are one, so no cancellation is possible. None of the three complete rational preimages lies in $\mathbb{Z}[S^{-1}]^3$. Local integrality at every prime was proved for every n in theorem 3.1. $\square$

For example, $n = 6$ gives the target $(0, -60, 2)$ and denominators $-30, 66, 55$. None is a power of two up to sign, so this example has no $\mathbb{Z}[1/2]$ -point. The obstruction is therefore not a defect caused solely by bad reduction at 2.

Corollary 5.2 (The same dense obstruction over every fixed S). *For a fixed finite set S , let $M_S(H)$ count the parameter pairs in theorem 4.3 whose target has no preimage over $\mathbb{Z}[S^{-1}]$. Then*

$$M_S(H) = \frac{27}{4\pi^2} (2H + 1)^2 + O_S(H \log H).$$

Their targets are Zariski dense in $C = 2$ and have integral preimages over $\mathbb{Z}_p$ for every prime, including the primes of S .

Proof. All second and third coordinates in (4.2) are integers. A preimage belongs to $\mathbb{Z}[S^{-1}]^3$ exactly when some D_r has no prime factor outside S . Then both nonzero integer differences $r - s$ and $r - t$ have all prime factors in S as well.

In the parameter box, all differences have absolute value at most $3H + 1$. The number of signed nonzero integers of this size supported on S is at most

$$2 \prod_{p \in S} \left(1 + \left\lfloor \frac{\log(3H + 1)}{\log p} \right\rfloor \right) = O_S((\log H)^{|S|}).$$

Given two such differences $u = r - s$, $v = r - t$, the root sum forces $3r = 1 + u + v$ and hence determines the root triple, up to its finitely many orderings. Thus only $O_S((\log H)^{2|S|})$ parameter pairs in the whole box can give an S -integral point. Removing this set from the locally soluble pairs of theorem 4.3 preserves its stated asymptotic; for fixed S the polylogarithmic error is absorbed into $O_S(H \log H)$. Positive parameter density implies target Zariski density by the proof of corollary 4.4. The all-prime local assertion was not weakened when these pairs were removed. $\square$

5.2 Transport to the repository’s stabilized maps

Proposition 5.3 (Integral stable-equivalence invariance). *Let R be either $\mathbb{Z}$ or $\mathbb{Z}[S^{-1}]$, let $f \in R[X]^d$, and let*

$$h = \beta \circ (f \times \text{id}_{\mathbb{A}^m}) \circ \alpha,$$

where α, β are polynomial automorphisms over R with polynomial inverses over R . For a target $\mathbf{b} \in R^d$, put $\mathbf{b}' = \beta(\mathbf{b}, 0)$. Over every R -algebra T there is a bijection

$$h^{-1}(\mathbf{b}')(T) \simeq f^{-1}(\mathbf{b})(T).$$

Consequently integral local–global failures transport to h over R .

Proof. Apply β^{-1} to the target equation and then α to a source point. The equation becomes $(f \times \text{id})(v, u) = (\mathbf{b}, 0)$, which forces $u = 0$ and $f(v) = \mathbf{b}$. Each step is invertible over every T because all polynomial coefficients and inverse coefficients lie in R . $\square$

The source and target shears in ProveIt’s stable degree reductions are of this form [2]. Thus the arithmetic obstruction is not tied to the expanded degree-seven presentation. It transfers to each such integral stable representative. The proposition is unconditional; when applied to a particular repository reduction, the equality defining that reduction retains the proof status documented by the repository. We do not upgrade its symbolic certificates to kernel-checked proofs.

5.3 The exact profinite closure

Theorem 5.4 (Closure of the integral image). *For any polynomial map $f \in \mathbb{Z}[X_1, \dots, X_d]^r$,*

$$\overline{f(\mathbb{Z}^d)}^{\widehat{\mathbb{Z}}^r} = f(\widehat{\mathbb{Z}}^d) = \prod_p f(\mathbb{Z}_p^d). \quad (5.1)$$

For the map F , the difference between this closure and $F(\mathbb{Z}^3)$, intersected with $\mathbb{Z}^3$, contains every $\mathbf{b}_n$ and is Zariski dense in the target plane $C = 2$.

Proof. The profinite integers $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ are compact, and polynomial evaluation is continuous. Hence $f(\widehat{\mathbb{Z}}^d)$ is closed. The diagonal image of $\mathbb{Z}^d$ is dense by the Chinese remainder theorem, so its image is dense in the compact image: every neighborhood of $f(x)$ pulls back to a neighborhood of x meeting $\mathbb{Z}^d$. This proves the first equality. Polynomial evaluation on the product is coordinatewise, and choices of preimages can be made separately at each prime, proving the second equality. The final assertions follow from [theorem 3.1](#) and [corollary 4.4](#). $\square$

Thus even *all* congruences, not merely all congruences below a fixed bound, fail to characterize the integer image. On a split fiber, [lemma 3.3](#) identifies exactly what they miss: local points can live on different rational components.

6 The contrasting rational Hasse principle

6.1 A cubic obstruction at infinitely many primes

We use one standard external theorem in this section: Chebotarev’s density theorem. In the form needed here, every conjugacy class in the Galois group of a finite Galois extension of number fields occurs as Frobenius at infinitely many unramified finite places. The factorization of a separable polynomial at a good place is given by the cycle lengths of that Frobenius action. See Milne [6, Theorem 8.31 and Example 8.36]. No effective version is required.

Lemma 6.1. *If an irreducible cubic over a number field K has a root in every finite completion K_v , a contradiction follows. In fact, it has no root at infinitely many finite places.*

Proof. Its Galois group acts transitively on three roots and is either the cyclic group of order three or the full symmetric group S_3 . In either case it contains a three-cycle. Chebotarev supplies infinitely many good places with that Frobenius cycle type. The reduction is an irreducible cubic there. After excluding the finitely many places where its monic normalization is not integral or its discriminant is not a unit, a local root would be integral and would reduce to a root of this irreducible cubic. This is impossible. $\square$

Theorem 6.2 (Rational local–global equality). *For every number field K and $\mathbf{b} \in K^3$,*

$$\mathbf{b} \in F(K^3) \iff \mathbf{b} \in F(K_v^3) \text{ for every finite place } v. \quad (6.1)$$

Archimedean places can be added without changing the equality.

Proof. The forward implication is immediate. Write $\mathbf{b} = (A, B, C)$ and assume local solubility at every finite place. If $C = 0$, the point $(0, B, A - 4B^2)$ lies in K^3 . Suppose $C \neq 0$. By [theorem 2.2](#), local solubility means that $g_{A,B,C}$ has a simple root in every K_v .

An irreducible g is impossible by [lemma 6.1](#). A reducible cubic has a root in K . If some K -root is simple, reconstruction finishes the proof. If a K -root has multiplicity two, the remaining linear factor is also defined over K and is simple. The only remaining possibility is a triple root, in which case no simple root exists even geometrically and every local fiber is empty. This contradicts the assumption. Thus a simple K -root always exists. $\square$

Remark 6.3 (Reconciliation with the integral counterexamples). The targets $\mathbf{b}_n$ do have rational preimages, so they satisfy [theorem 6.2](#). What fails is simultaneous integrality of one of those points. The earlier degree-five construction with no rational preimage [5] is of a different kind: a product of an irreducible cubic and its quadratic discriminant-field polynomial can have a root at every completion without a rational root. A single cubic cannot realize that covering mechanism. We claim neither a new general Hasse theorem for arbitrary polynomial maps nor a contradiction to the known degree-five examples.

6.2 A practical exact rational inverse algorithm

The proof gives a short, complete rational algorithm. On input $(A, B, C) \in \mathbb{Q}^3$, factor $g_{A,B,C}$ over $\mathbb{Q}$. Retain each rational root of multiplicity one, reconstruct it by (2.6), and add (2.7) when $C = 0$. To decide integer or S -integer membership, inspect the reduced denominators of this finite list. An irreducible quadratic factor can be ignored for the rational list, but not for a local integral-image problem.

The file `code/rational_inverse.py` implements this procedure using exact rational arithmetic and SymPy’s rational factorization. It verifies every reconstructed point by substitution. The mathematical completeness of the procedure is [theorem 2.2](#), rather than any heuristic search bound.

7 Known finite-field laws and their arithmetic use

7.1 Odd-characteristic fiber counts

For a finite field $k = \mathbb{F}_q$ of odd characteristic, let

$$N_j(q) = \#\{\mathbf{b} \in k^3 : \#F^{-1}(\mathbf{b})(k) = j\}.$$

The following formulas are prior results from [4]. We include their elementary proof to fix exactly which roots are counted.

Theorem 7.1 (Finite-field law, credited to the archived note). *If char $k > 3$, the only fiber sizes are 0, 1, 3, and*

$$N_0 = \frac{(q-1)(q^2+2)}{3}, \quad N_1 = \frac{q^3+q^2-2q+2}{2}, \quad N_3 = \frac{(q-1)(q^2+2)}{6}. \quad (7.1)$$

If char $k = 3$, the only fiber sizes are again 0, 1, 3, but

$$N_0 = \frac{q^2(q-1)}{3}, \quad N_1 = \frac{q^2(q+1)}{2}, \quad N_3 = \frac{q^2(q-1)}{6}. \quad (7.2)$$

Proof. When $C \neq 0$, a cubic has 0, 1, or 3 simple roots in its ground field: two distinct simple rational roots force the third root to be rational and simple. A repeated-root cubic has at most one simple root. When $C = 0$, the boundary point is always present; its affine quadratic has either zero or two simple rational roots. This gives the same list of possible fiber sizes.

For $C \neq 0$, a three-point fiber is determined by an unordered triple of distinct roots $\{r, s, t\}$ with nonzero sum. The sum determines $C = 2/(r + s + t)$, and the other elementary symmetric functions determine B, A . For $C = 0$, a three-point fiber is determined by an unordered pair of distinct roots of $-2T^2 + BT - 2A$. Therefore

$$N_3 = \binom{q}{2} + \binom{q}{3} - Z_q, \quad (7.3)$$

where Z_q counts unordered distinct zero-sum triples.

There are q^2 ordered zero-sum triples in total. In characteristic not three, each equality event $r = s$, $s = t$, or $t = r$ has q solutions; all their pairwise and triple intersections consist only of $(0, 0, 0)$. The union has $3q - 2$ elements. Hence $6Z_q = q^2 - 3q + 2 = (q-1)(q-2)$. In characteristic three, any equality in a zero-sum triple forces all three entries equal, and there are q such triples. Thus $6Z_q = q^2 - q$ instead. Substitution in (7.3) gives both values of N_3 .

Finally, counting targets and source points respectively gives $q^3 = N_0 + N_1 + N_3$ and $q^3 = N_1 + 3N_3$. Thus $N_0 = 2N_3$ and $N_1 = q^3 - 3N_3$, proving every formula. $\square$

7.2 A fixed-leading-coefficient refinement

We will need one more known count from [4], proved here independently by trace counting. Fix $C \neq 0$ and assume char $k > 3$. As (A, B) varies, g/C ranges over all monic cubics with the fixed coefficient $-2/C$ of T^2 . The irreducible ones number

$$\frac{q^2 - 1}{3}. \quad (7.4)$$

Indeed, the trace map $\mathbb{F}_{q^3} \rightarrow \mathbb{F}_q$ has q^2 elements of any prescribed trace. Exactly one of those lies in $\mathbb{F}_q$, since the trace of a ground-field element is three times that element. Every other element has degree three and belongs to a Frobenius orbit of size three. This proves (7.4). Across all $C \neq 0$, the number of target residue classes whose inverse binary cubic has *no* projective root is consequently

$$I(q) = \frac{(q-1)(q^2-1)}{3} = \frac{(q-1)^2(q+1)}{3}. \quad (7.5)$$

These classes form a useful congruence obstruction to rational solubility. They should not be confused with *all* omitted targets of F over k , because a repeated projective root is still a root but contributes no finite source point.

8 An inverse-ball theorem at every precision

8.1 The exact image of a sufficiently small ball

We now establish the analytic bridge which makes a finite dyadic calculation conclusive. The lemma is a quantitative version of the p -adic inverse-function argument, with its image lattice retained explicitly. We give a proof rather than relying on a convergence claim.

Lemma 8.1 (Exact inverse ball). *Let $f : \mathbb{Z}_p^d \rightarrow \mathbb{Z}_p^d$ be given by polynomials with coefficients in $\mathbb{Z}_p$, let $a \in \mathbb{Z}_p^d$, and set $J = Df(a)$. Suppose $v_p(\det J) = e < \infty$. For every integer $k > e$, the restriction of f to $a + p^k \mathbb{Z}_p^d$ is a bijection onto*

$$f(a) + p^k J \mathbb{Z}_p^d. \quad (8.1)$$

This image is a union of target residue classes modulo p^{k+e} . For every $m \geq k + e$, a target residue class modulo p^m has either zero or exactly p^e preimages modulo p^m within the source class $a \pmod{p^k}$. It has p^e precisely when its lifts lie in (8.1).

Proof. Taylor expansion, with integral polynomial coefficients, gives

$$f(a + p^k t) = f(a) + p^k J t + p^{2k} R(t),$$

where $R : \mathbb{Z}_p^d \rightarrow \mathbb{Z}_p^d$ is an integral polynomial map and is 1-Lipschitz for the maximum norm. The adjugate formula shows that $p^e J^{-1}$ is integral. Define

$$\Psi(t) = t + p^k J^{-1} R(t).$$

Its correction term is Lipschitz with constant at most $p^{-(k-e)} < 1$. For every $v \in \mathbb{Z}_p^d$, the fixed-point iteration

$$t \mapsto v - p^k J^{-1} R(t)$$

is a contraction of the complete space $\mathbb{Z}_p^d$ into itself. It has a unique fixed point, proving that Ψ is bijective. The same strict inequality gives $\|\Psi(t) - \Psi(t')\|_p = \|t - t'\|_p$, so Ψ induces a bijection modulo every p^r . Since

$$f(a + p^k t) = f(a) + p^k J \Psi(t),$$

we have both injectivity and the exact image (8.1).

Also $p^e \mathbb{Z}_p^d \subset J \mathbb{Z}_p^d$, so the image contains a full target ball of radius $p^{-(k+e)}$ around each of its points. It is therefore a union of residue classes modulo p^{k+e} .

For the finite count, the Smith normal form of J has diagonal entries with valuations $e_1, \dots, e_d \geq 0$ and $\sum e_i = e$. With $r = m - k \geq e$, the congruence $Jt \equiv v \pmod{p^r}$ has $p^{e_1 + \dots + e_d} = p^e$ solutions whenever it is soluble. The bijection induced by Ψ leaves that count unchanged. Finally, $p^m \mathbb{Z}_p^d \subset p^k J \mathbb{Z}_p^d$ for $m \geq k + e$, so solubility modulo p^m is equivalent to membership in the actual image lattice. This proves the last assertion without extrapolating from finite precision. $\square$

Corollary 8.2 (Uniform determinant valuation). *Suppose $v_p(\det Df(a)) = e$ for every $a \in \mathbb{Z}_p^d$, and fix $k > e$. Then the actual integral fiber cardinality $J_f(\mathbf{b}) = \#f^{-1}(\mathbf{b})(\mathbb{Z}_p)$ is finite and determined by $\mathbf{b} \pmod{p^{k+e}}$. For every $m \geq k + e$,*

$$\#\{a \pmod{p^m} : f(a) \equiv \mathbf{b} \pmod{p^m}\} = p^e J_f(\mathbf{b}). \quad (8.2)$$

In particular, both the image and its fiber-cardinality strata are clopen.

Proof. Partition the source into its p^{kd} residue balls modulo p^k . Each ball contributes either one actual preimage or none, and its image membership depends only on the target class modulo p^{k+e} by lemma 8.1. Sum the point counts and the finite congruence counts over these disjoint source balls. $\square$

8.2 The good-prime consequences

For F , every odd prime has $e = 0$, so we can take $k = 1$. Each source point modulo p lifts uniquely above every target lift with that residue. Thus, exactly and not just asymptotically,

$$\Pr_{\mathbf{b} \in \mathbb{Z}_p^3} (\#F^{-1}(\mathbf{b})(\mathbb{Z}_p) = j) = \frac{N_j(p)}{p^3}, \quad p \text{ odd}, j = 0, 1, 3. \quad (8.3)$$

This is the direct p -adic consequence of the previously known finite-field formulas. In particular,

$$\rho_p := \text{vol}(F(\mathbb{Z}_p^3)) = \begin{cases} 7/9, & p = 3, \\ \frac{2}{3} + \frac{1}{3p} - \frac{2}{3p^2} + \frac{2}{3p^3}, & p > 3. \end{cases} \quad (8.4)$$

The same lifting proof works over the ring of integers of an unramified extension of $\mathbb{Q}_p$, with p replaced in the finite-field counts by its residue cardinality. We do not use that extension below.

9 The exceptional prime: the exact dyadic law

9.1 From 64 source balls to 512 target classes

At $p = 2$, the determinant has constant valuation one. Taking $e = 1$, $k = 2$ in [lemma 8.1](#) partitions the source into 64 balls $a + 4\mathbb{Z}_2^3$. Their images are

$$F(a) + 4J_F(a)\mathbb{Z}_2^3. \quad (9.1)$$

Because $v_2(\det J_F(a)) = 1$, the matrix $J_F(a) \bmod 2$ has rank two. Thus each image in (9.1) consists of exactly four classes modulo 8, specifically

$$\{F(a) + 4J_F(a)h \pmod{8} : h \in \{0, 1\}^3\}. \quad (9.2)$$

Every target in such an image has exactly one preimage in that source ball. Therefore the desired integral fiber cardinality is obtained by counting how many of the 64 four-element sets contain the target class.

Theorem 9.1 (Exact dyadic integral law). *For the fixed map F in (1.1), the number $j(\mathbf{b}) = \#F^{-1}(\mathbf{b})(\mathbb{Z}_2)$ is determined by $\mathbf{b} \bmod 8$. Its Haar distribution is*

$$\begin{array}{c|cccc} j & 0 & 1 & 2 & 3 \\ \hline \Pr(j(\mathbf{b}) = j) & 21/32 & 7/32 & 3/32 & 1/32. \end{array} \quad (9.3)$$

In particular,

$$\text{vol}(F(\mathbb{Z}_2^3)) = \frac{11}{32}. \quad (9.4)$$

For every $m \geq 3$, the possible fiber sizes of $F : (\mathbb{Z}/2^m\mathbb{Z})^3 \rightarrow (\mathbb{Z}/2^m\mathbb{Z})^3$ are 0, 2, 4, 6, with counts

$$2^{3m} \left(\frac{21}{32}, \frac{7}{32}, \frac{3}{32}, \frac{1}{32} \right) \quad (9.5)$$

respectively. Modulo 8 is necessary for image membership: modulo 4 does not suffice.

Proof. The reduction to (9.2) proves that a finite table at modulus 8 settles the infinite-precision assertion. The exact table is below. Each row fixes the parity of (A, B, C) and thus contains 64 target classes modulo 8.

$(A, B, C) \bmod 2$	$j = 0$	$j = 1$	$j = 2$	$j = 3$
(0, 0, 0)	0	48	0	16
(0, 0, 1)	32	32	0	0
(0, 1, 0)	48	0	16	0
(0, 1, 1)	64	0	0	0
(1, 0, 0)	32	32	0	0
(1, 0, 1)	64	0	0	0
(1, 1, 0)	48	0	16	0
(1, 1, 1)	48	0	16	0
Total	336	112	48	16

Here is a complete finite enumeration defining and checking the total row using only integer arithmetic. The derivative is not needed in this version: [corollary 8.2](#) says that each actual preimage contributes exactly two source residues modulo 8.

```

from collections import Counter
from itertools import product

def F(x, y, z):
    u = 1 + x*y
    h = u*u*z + y*y*(1 + 3*u)
    return u*h, y + 3*x*h, x*(5 - 3*u - x*x*z)

count = Counter(tuple(v % 8 for v in F(*a))
                 for a in product(range(8), repeat=3))
assert all(c % 2 == 0 for c in count.values())
actual = {b: count[b] // 2
          for b in product(range(8), repeat=3)}
assert Counter(actual.values()) == {0:336, 1:112, 2:48, 3:16}

```

The accompanying certificate independently constructs all 64 sets in (9.2) using the product-rule Jacobian and verifies equality *for every target residue*, not merely agreement of the total histogram. It also reproduces each parity row. This finite integer enumeration is the computational part of the proof; the all-precision part is [lemma 8.1](#).

Divide the total row by 512 to obtain (9.3) and sum its nonzero columns to obtain (9.4). Equation (8.2) multiplies actual cardinalities by two at every modulus 2^m , $m \geq 3$, proving (9.5). Finally, the target classes (0, 1, 0) and (0, 1, 4) agree modulo 4; the former has two integral 2-adic preimages and the latter none, as certified by the same pointwise table. Hence image membership cannot be determined modulo 4. $\square$

9.2 A two-point integral fiber and the importance of the domain

The target (0, 1, 0) has three rational preimages:

$$(0, 1, -4), \quad (2, -1/2, 5/4), \quad (-2, 1, 2).$$

Only the first and third are in $\mathbb{Z}_2^3$, so its dyadic integral fiber has two points. This is compatible with the three-point rational fiber and the known geometric law. In fact, it also has exactly two ordinary integer preimages.

Remark 9.2 ([write] An intake observation: the project’s collision fiber). This remark is *not* a claim of the manuscript. It was observed when the manuscript was taken into ProveIt and checked there by exact

computation (SymPy 1.14.0: evaluation, and a lex Gröbner basis whose x -polynomial is $x(x-1)(x+1)$). Apply [theorem 2.2](#) to the project’s collision value $(0, -2, 0)$ (`collisionValue`, [section 1.3](#)). Since $C = 0$ there is the boundary point $(0, -2, -16)$, and $g(T) = -2T(T+1)$ has the simple roots $t = 0$ and $t = -1$, which reconstruct $(-1, 1, 5)$ and $(1, -2, 8)$. So

$$F^{-1}(0, -2, 0)(\overline{\mathbb{Q}}) = \{(-1, 1, 5), (0, -2, -16), (1, -2, 8)\},$$

three *integral* points: F has an integral triple collision. The project records and formalizes only the first two (`collision0`, `collision1`). Along the project’s family, whose parameter (called t in the project README) is written τ here to keep it apart from the root $t = y + 1/x$, $F(\tau, -1/\tau, 5/\tau^2) = F(0, 2/\tau, -16/\tau^2) = (0, 2/\tau, 0)$ and the third point is $(-\tau, 2/\tau, 8/\tau^2)$. The torus action $F(x/s, sy, s^2z) = (s^2P, sQ, R/s)$ at $s = -\frac{1}{2}$ carries these three points to $(2, -\frac{1}{2}, \frac{5}{4})$, $(0, 1, -4)$ and $(-2, 1, 2)$, the fiber over $(0, 1, 0)$ displayed above; its two integral points have largest coordinate 4, against 16 for the project’s integral collision.

There are three different distributions that must not be interchanged:

Source and target	Fiber sizes	Image fraction or measure
$\mathbb{F}_2^3$	0, 1, 3	$6/8 = 3/4$
$(\mathbb{Z}/4\mathbb{Z})^3$	0, 2, 6	$28/64 = 7/16$
$\mathbb{Z}_2^3$	0, 1, 2, 3	$11/32$
$(\mathbb{Z}/2^m\mathbb{Z})^3, m \geq 3$	0, 2, 4, 6	$11/32$

The actual modular histograms for moduli 2, 4, 8 are respectively

$$(N_0, N_1, N_3) = (2, 5, 1), \quad (N_0, N_2, N_6) = (36, 26, 2), \quad (N_0, N_2, N_4, N_6) = (336, 112, 48, 16).$$

Also, passing to a finite field $\mathbb{F}_{2^r}$ is not the same operation as passing to the ring $\mathbb{Z}/2^r\mathbb{Z}$. The characteristic-two finite-field formulas in [\[4\]](#) do not by themselves imply the dyadic law.

As a consistency check, the expected integral dyadic fiber size is

$$\frac{7}{32} + 2\frac{3}{32} + 3\frac{1}{32} = \frac{1}{2}.$$

This also follows from summing the contributions of 64 source balls, each with an image of measure $1/128$. The corresponding expected fiber size at an odd prime is one. These are exactly the factors suggested by the determinant’s local absolute value, but the argument here is a finite lattice count rather than an appeal to a change-of-variables formula.

10 Sparse rational images and a measure-zero congruence closure

10.1 A closed subset of profinite measure zero

Theorem 10.1. *The compact image $F(\widehat{\mathbb{Z}}^3)$ has Haar measure zero in $\widehat{\mathbb{Z}}^3$. Nevertheless, its intersection with $\mathbb{Z}^3$ contains the Zariski-dense split-plane family of [corollary 4.4](#).*

Proof. By [theorem 5.4](#), the image is the product of the local images. For $p > 3$, [\(8.4\)](#) gives $\rho_p < 3/4$. The image is contained in the cylinder imposing local membership at any chosen N primes greater than three, and that cylinder has measure at most $(3/4)^N$. Letting N tend to infinity proves measure zero. The second assertion was already proved in [theorem 5.4](#). $\square$

This is a density statement in a compact topological space. It neither asserts emptiness nor contradicts Zariski density in a plane. Topological measure, polynomial equations, and integer counting measure detect different notions of size.

10.2 A quantitative bound for rationally soluble integral targets

Let

$$\mathcal{A}(H) = \#(F(\mathbb{Q}^3) \cap [-H, H]^3 \cap \mathbb{Z}^3), \quad H \geq 1.$$

Unlike an integer-image sieve, a sieve for $F(\mathbb{Q}^3)$ cannot simply exclude all residue classes outside $F(\mathbb{F}_p^3)$: a rational source may have a p -adic denominator. The correct necessary reduction condition is that the *binary* inverse cubic have a projective root modulo p .

Lemma 10.2. *If $(A, B, C) \in \mathbb{Z}^3 \cap F(\mathbb{Q}^3)$, then $G_{A,B,C}$ has a projective root over $\mathbb{F}_p$ for every prime $p > 2$.*

Proof. The rational source gives a rational projective root by [theorem 2.2](#). Represent it by coprime integers (T, S) . At every prime they are not both zero after reduction, and the homogeneous equation remains valid. The coefficient -2 ensures that the reduced binary polynomial is not the zero polynomial for odd p . The reduced root need not remain simple, which is why the stronger exclusion would be invalid. $\square$

Theorem 10.3 (An elementary quantitative sieve). *Let $p_1, \dots, p_N$ be distinct primes greater than three and $M = \prod_{i=1}^N p_i$. Then*

$$\mathcal{A}(H) \leq \left(\prod_{i=1}^N \alpha_{p_i} \right) (2H + 1 + M)^3, \quad \alpha_p = \frac{2}{3} + \frac{1}{3p} + \frac{1}{3p^2} - \frac{1}{3p^3} < \frac{3}{4}. \quad (10.1)$$

In particular, $F(\mathbb{Q}^3) \cap \mathbb{Z}^3$ has zero density in integer boxes. There is an absolute $c > 0$ such that

$$\mathcal{A}(H) \ll H^3 \exp(-c\sqrt{\log H}). \quad (10.2)$$

Proof. By [\(7.5\)](#) and [lemma 10.2](#), a fraction $I(p)/p^3$ of target residues is forbidden at p . The allowed fraction is exactly α_p . The Chinese remainder theorem makes the number of allowed classes modulo M equal to $M^3 \prod_i \alpha_{p_i}$. Each class contains at most $((2H + 1)/M + 1)^3$ points of the integer cube. This proves [\(10.1\)](#). For $p \geq 5$, the positive correction $1/(3p) + 1/(3p^2)$ is at most $2/25 < 1/12$, proving $\alpha_p < 3/4$.

For density zero, first fix N , divide by $(2H + 1)^3$, and let $H \rightarrow \infty$. The upper limit is at most $(3/4)^N$; then let $N \rightarrow \infty$. For the stated quantitative estimate, use the elementary Bertrand bound that between n and $2n$ there is a prime for every integer $n > 1$. Taking the primes in increasing order starting at five gives $p_i < 2^{i+2}$, and hence

$$M < 2^{N(N+5)/2}.$$

Choose the largest nonnegative integer N with $N(N+5)/2 \leq \log_2 H$. Then $M \leq H$ and $N = \sqrt{2 \log H / \log 2} + O(1)$. Substituting into [\(10.1\)](#) gives

$$\mathcal{A}(H) \ll H^3 \exp \left(-\log(4/3) \sqrt{\frac{2}{\log 2}} \sqrt{\log H} \right),$$

with an absolute multiplicative constant, proving the claim. Only the quantitative refinement uses Bertrand's theorem; the zero-density conclusion uses neither it nor Chebotarev. $\square$

This weak but explicit bound is not advertised as an optimal thin-set estimate. It is useful here because it is derived directly from the same inverse cubic and makes the necessary rational-versus-integral sieve correction transparent.

11 Verification artifacts and the route to formal proofs

11.1 What was actually executed

The accompanying archive contains three independent-purpose checkers and an exact rational inverse utility. They were run in the preparation of this article, with results recorded under `certificates/`. [write] In this report that directory is shipped as `data/`, with the same file names; two of the scripts still write to `../certificates/` next to their own directory ([section C.3](#)).

Symbolic identities. `verify_symbolic.py`, run with SymPy 1.14.0, verifies 27 exact identities. These include $\det JF = -2$, both inverse charts, the inverse cubic and derivative identities, the discriminant, the symbolic one-parameter factorization, all three symbolic family branches, and characteristic-three identities. Every zero test is a polynomial-numerator identity after exact rational simplification; no floating-point tolerance is involved.

Finite local certificates. `verify_finite.py` uses only the Python standard library. It checks factored and expanded polynomial evaluators, all three one-parameter branches for $2 \leq n \leq 100$ with exact fractions, the complete target-wise modulus-eight dyadic certificate in two ways, and stabilization for every target modulo 16 and 32. It also reproduces the known finite-field laws for primes from 3 through 31 and checks good-prime lifting at moduli 9, 27, 25. Finally, it constructs exact CRT preimages for four composite moduli. The finite range in these checks is not the theorem’s range; the universal conclusions are proved in the preceding sections.

The split plane. `verify_split_plane.py` uses only exact integers and fractions. It checks all 1600 distinct-root fibers with $(a, b) \in [-20, 20]^2$, including the exact denominator $\gcd g^2$. Its parameter counts through $H = 250$ provide a check on [theorem 4.3](#), not evidence substituted for the Möbius proof.

11.2 Trust boundaries

A successful Python or SymPy computation is not a Lean kernel proof. The new theorems in this article have not been compiled in Lean or Rocq, and no claim is made that they are already part of ProveIt’s trusted theorem boundary. The dyadic total is a finite, reproducible computer-assisted enumeration plus an ordinary mathematical proof of its all-precision significance. A fully kernel-checked version should verify that finite enumeration with a proved checker.

Conversely, the main integral-family and parameter-density theorems do not depend on computational discovery being trustworthy. Their proofs use explicit factorization, a complete inverse theorem, $\gcd$ identities, and elementary counting, all written out in the article. The scripts supply additional independent checks of the formulas.

11.3 A dependency-conscious formalization plan

A practical Lean extension should proceed in four layers.

Layer 1: exact algebra over a field. Reuse the repository’s definition of F . Formalize [\(2.3\)](#), [\(2.5\)](#), and [\(2.6\)](#) over a field with $2 \neq 0$. The central API should state both directions of the preimage characterization

and explicitly separate $x = 0$. Polynomial normalization proves the identities; the main logical obligation is completeness, not evaluation of examples.

Layer 2: rational denominators and local selection. Formalize [theorems 3.1](#) and [4.1](#) over $\mathbb{Q}$ and $\mathbb{Z}$. The no-integer point result needs only denominator arithmetic once completeness is available. Before introducing p -adic numbers, prove the stronger constructive congruence statement for every natural modulus by modular inverses and CRT. For each prime, the rational selected branch is already known explicitly; embedding it into $\mathbb{Q}_p$ should be a final transport step rather than a search inside a local field.

Layer 3: the finite dyadic certificate. Represent source and target residues as triples of $\mathbb{Z}\text{Mod } 8$, prove that the evaluator is reduction of the formal polynomial, and verify the 512-entry count. The semantic bridge is crucial: a theorem about an uninterpreted list of integers is insufficient. The independent 64-ball version can instead certify image incidence using the mod-two Jacobian. No global search or oracle is needed at this layer.

Layer 4: analysis and external arithmetic. Formalize the inverse-ball lemma using completeness and contraction, then connect Layer 3 to actual $\mathbb{Z}_2$ fibers. Haar measures of cylinders finish the distribution theorem. The rational Hasse theorem is a separate branch requiring a suitable Chebotarev theorem; it is not a dependency of the explicit integral counterexamples. The parameter density and thin sieve form further analytic-number-theory branches. Keeping these branches separate prevents a deep external result from becoming an unnecessary assumption of the elementary headline family.

12 Further research questions

The following questions are proposed research targets. They are not represented as settled results, and their priority status would require additional literature review before publication.

12.1 A minimal congruence description at the prime two

Research question 12.1. Can the 512-entry dyadic certificate be compressed into a short list of explicit congruence conditions on (A, B, C) , with a hand-checkable case proof for each integral fiber size?

[theorem 9.1](#) proves that modulus eight suffices and modulus four does not. Thus the remaining problem is logical and algebraic compression, not an unbounded precision search. The eight parity rows suggest a small branching structure. A compact description would improve both exposition and formal verification.

12.2 All locally soluble integral fibers, not just split integral roots

Research question 12.2. Classify the integral targets with three rational preimages, and then those with a single rational preimage and a quadratic pair, according to local integral solubility at every prime.

For arbitrary split rational fibers, [lemma 3.3](#) already gives a criterion once denominators are known. The challenge is to express it in target coefficients without factoring every fiber individually. The quadratic case is qualitatively different: the nonrational pair can supply local integral points at primes where the rational point fails integrality. A complete classification must record both factorization and valuations of the reconstruction derivative.

12.3 Target-height asymptotics for integral Hasse failures

Research question 12.3. What is the asymptotic number of locally integral but globally nonintegral targets on $C = 2$ in boxes $|A| \leq H_A$, $|B| \leq H_B$?

The density $27/(4\pi^2)$ counts ordered root parameters in a square. Passing to target heights introduces a nonlinear region, a sixfold symmetry away from the discriminant, and cusp-like ranges where one root is small. These are genuine counting issues. Establishing a main term with an explicit Euler product would sharpen the present Zariski-density result substantially.

12.4 Density in the entire three-dimensional target space

Research question 12.4. Does the set of integral Hasse failures for this fixed F form a Zariski-dense subset of all of $\mathbb{A}^3$, rather than just the plane $C = 2$?

The present construction does not answer this: all its targets have the same third coordinate. Rational weighted symmetries can move that coordinate, but they can introduce new prime denominators and do not preserve all local-integrality conditions automatically. A valid extension must control those exceptional primes explicitly.

12.5 Exact local laws over extensions of $\mathbb{Q}_2$

Research question 12.5. Determine integral fiber distributions over finite extensions of $\mathbb{Q}_2$, including ramified extensions.

Over a discretely valued extension, the same contraction proof uses the valuation of 2 in that field. It therefore gives a finite precision bound, but the number of residue classes and the rank patterns can change. It would be particularly useful to find a uniform formula in residue cardinality and ramification index, rather than a new exhaustive table for each extension.

12.6 Effective local witnesses for rational insolubility

Research question 12.6. For an integral target with irreducible inverse cubic, how small can one guarantee a prime p with no $\mathbb{Q}_p$ -preimage, in terms of the target's height and cubic discriminant?

[theorem 6.2](#) uses qualitative Chebotarev. An effective specialized estimate could turn the abstract rational Hasse principle into a practical proof-producing negative decision procedure. Unconditional and conditional bounds should be separated, and the finite exceptional set where the inverse polynomial is not an integral separable cubic must be included in the implementation.

12.7 Finer image-density estimates

Research question 12.7. Can one obtain the correct exponent or a main-term asymptotic for $\mathcal{A}(H)$, rather than the elementary bound in [theorem 10.3](#)?

The projective root condition gives a uniform positive congruence loss, but the simple CRT argument uses the moduli inefficiently. A large-sieve approach or a direct height analysis of rational roots might yield a substantially stronger bound. Such a result would concern the arithmetic image of a concrete nonproper étale map, not merely its geometric degree.

12.8 Minimal arithmetic complexity under stable equivalence

Research question 12.8. Among the integral stable representatives in ProveIt, which realizes integral Hasse failures with the smallest target height, source denominator height, degree, or ambient dimension?

[proposition 5.3](#) guarantees existence but not small formulas after transport. Complexity should be reported as a vector of separate measures; minimizing degree alone can raise dimension and coefficient height. A search accompanied by exact certificates could produce useful representatives without claiming a global optimum from a bounded search.

12.9 A reusable formal library for finite arithmetic fibers

Research question 12.9. Can the combination of a complete inverse certificate, denominator covering, and the inverse-ball theorem become a generic proof-producing workflow for polynomial maps with small geometric fibers?

The division of labor in [section 11](#) suggests a reusable architecture: exact elimination produces a candidate inverse; algebraic lemmas certify completeness; finite local tables are reflected; and an analytic theorem upgrades them to all precisions. This would convert many computational observations into theorems with an explicit trust boundary, rather than a collection of map-specific numerical checks.

13 Conclusion

The original polynomial witness raises an arithmetic question quite different from noninjectivity. For this particular map, rational local solubility everywhere does force a rational point, but integral local solubility everywhere does *not* force an integral point. The failure occurs in explicitly split three-point fibers, survives inversion of any fixed finite set of primes, and fills a Zariski-dense subset of a target plane. On that split-root parameter plane its exact density is $27/(4\pi^2)$.

The local side is also completely explicit at the exceptional prime: modulus eight determines the integral dyadic image and every integral fiber size, with image measure $11/32$. This is an arithmetic refinement of the existing cubic model, not a replacement for the known geometric or finite-field results. The general inverse-ball argument explains why a finite certificate can settle the infinite local question.

These results provide concrete new targets for ProveIt’s formalization program while keeping their status distinct: explicit mathematical proofs and replayed exact computations are supplied here; new kernel-checked formal proofs remain to be implemented.

A Replay guide and artifact inventory

The archive is self-contained except for standard TeX packages and the optional SymPy dependency. From its root directory:

```
python3 code/verify_finite.py --output certificates/finite_checks.json
python3 code/verify_split_plane.py
python3 code/verify_symbolic.py
python3 code/rational_inverse.py 0 -4 2
pdflatex -interaction=nonstopmode -halt-on-error arithmetic_fibers.tex
pdflatex -interaction=nonstopmode -halt-on-error arithmetic_fibers.tex
```

The first two commands use only Python’s standard library. The last two Python utilities need SymPy; the executed symbolic version was 1.14.0. The PDF build uses ordinary packages supplied in a full TeX Live installation, and does not require shell escape or external figures.

Path	Purpose
<code>arithmetic_fibers.tex</code>	Complete article source, bibliography included
<code>arithmetic_fibers.pdf</code>	Compiled article
<code>code/verify_finite.py</code>	Exact local tables, CRT construction, finite checks
<code>code/verify_split_plane.py</code>	Split-plane gcd and parameter counts
<code>code/verify_symbolic.py</code>	Universal algebraic identity checks
<code>code/rational_inverse.py</code>	Complete rational inverse utility
<code>certificates/*.json</code>	Machine-readable executed results
<code>certificates/*.txt</code>	Human-readable execution transcripts
<code>README.md</code>	Build instructions and statement-status summary
<code>SOURCE_REVIEW.md</code>	Pinned source paths and novelty boundaries

No external article PDFs, font files, or copies of third-party research repositories are redistributed in this archive.

[write] In ProveIt the article is `article.tex` (this text, rewritten as described in [section C](#)), `article.pdf` is a build of it, `certificates/` is shipped as `data/`, the delivered README is replaced by the report README, and the delivered PDF `arithmetic_fibers.pdf` is not shipped. The commands above are the delivery’s; [section C.3](#) gives commands that work with the shipped names.

B A characteristic-three simplification

The separate characteristic-three finite-field count has a direct algebraic explanation. Reducing (1.1) modulo 3 gives

$$Q = y, \quad P = z + y^2 - y^3 R, \quad R = -x - x^3 z.$$

For a target (A, B, C) , set $\zeta = A - B^2 + B^3 C$. The first two equations force $y = B$, $z = \zeta$; the remaining equation is

$$\zeta x^3 + x + C = 0.$$

Its derivative with respect to x is one. If $\zeta = 0$, it has exactly one geometric root; if $\zeta \neq 0$, it has three distinct geometric roots. Over a finite field its solution set is either empty or a translate of the kernel of the additive map $x \mapsto \zeta x^3 + x$, whose kernel has one or three elements. This confirms the possible rational fiber sizes 0, 1, 3, without carrying characteristic-zero cusp formulas through an invalid division by three. It is an explanatory calculation, not a new finite-field histogram claim.

C Provenance of this report

C.1 One manuscript

[write] This report is built from *one* manuscript: manuscript 03 of batch 36 of ProveIt’s incoming reports, the archive `ProveIt_Arithmetic_Local_Global` (inner directory `proveit_arithmetic_fibers`, main file `arithmetic_fibers.tex`, a 26-page PDF dated 24 September 2026, author line “Research draft prepared for Vladimir Reshetnikov”, PDF metadata “Research draft prepared with ChatGPT”), pinned to ProveIt

commit e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9. The archive arrived in commit e13affd32 and was placed in 1a1396d4d as a new report. Nothing was merged and nothing was selected away: every result, proof, example, table, question and limitation of the manuscript is printed. The manuscript contributed everything except the text marked [write].

The editorial changes are these.

- (1) Every label carries the prefix `alg:`. The source’s 77 labels are kept, unchanged after the prefix. Six were added: `alg:sub:attribution`, `alg:rem:collisionfiber`, `alg:app:report`, `alg:app:onesource`, `alg:app:pinned` and `alg:app:rerun`.
- (2) Attribution was made precise: a bracketed [write] clause in the abstract, the status note on the title page, and [section 1.3](#), which names the formalized statements of the project and says where the complex stratification and the finite-field laws come from.
- (3) [Remark 9.2](#) records an observation made at the write, not a claim of the manuscript. It is the only new numbered item and the last one of its section, so no source number moved.
- (4) Notes marked [write] were added in [section 11](#) and in [Appendix A](#) (shipped names), and this appendix is new.

No mathematical statement of the manuscript was changed and no symbol was renamed. The manuscript’s $t = y + 1/x$ is not the project’s t (which is xy in `Algebra/JacobianConjecture/Research/README.md` and the family parameter in the project `README`, which [remark 9.2](#) writes τ), and its p, q are a prime and a field size, not the project’s $p = xy^2, q = x^2yz$. The PDF bookmark of the title page was fixed (no duplicate page anchor), and its top space was reduced from 0.65 to 0.3 inch so that the added status note fits.

C.2 Repository statements at the pin

[write] The source read the root `README`, the project `README` and `Algebra/JacobianConjecture/Research/README.md` at its pin (its delivered `SOURCE_REVIEW.md`). Between the pin and the placement commit the project did not change, so its statements about ProveIt are current. They are accurate: the formal developments, the integral stable shears of [proposition 5.3](#) (unitriangular source and target maps over $\mathbb{Z}$), and the refusal to upgrade the project’s SymPy frontier certificates to kernel proofs. The manuscript claims no gap in the project; the question (1.2) is its own. The two external sources were checked at the write ([section 1.3](#)); Milne was not.

C.3 Rerunning the checks

[write] Rerun on a copy with the delivered layout, never in the report: `verify_split_plane.py` and `verify_symbolic.py` always write `split_plane_checks.json` and `symbolic_checks.json` into `../certificates/` relative to their own directory, and fail after passing their checks if it does not exist.

```
mkdir <scratch>          # outside the repository
cp -r code <scratch>/code
cp -r data <scratch>/certificates
cd <scratch>
python code/verify_finite.py --output certificates/finite_checks.json
python code/verify_split_plane.py
python code/verify_symbolic.py
python code/rational_inverse.py 0 -4 2
```

The last two need SymPy (the recorded runs used 1.14.0 and Python 3.13.5; no requirements file was delivered). At the write the four commands were run so with Python 3.13.5 and SymPy 1.14.0: every regenerated file equals the shipped one apart from Windows line endings. `data/build_review.json` is a static build record, not regenerated.

References

- [1] Vladimir Reshetnikov, *ProveIt: Jacobian conjecture project*, Algebra/JacobianConjecture/README.md and associated Lean/Rocq sources. Repository snapshot e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9, inspected September 24, 2026. [Pinned project documentation](#).
- [2] Vladimir Reshetnikov, *Searching for a simpler counterexample*, Algebra/JacobianConjecture/Research/README.md, in ProveIt, same pinned snapshot. [Pinned research documentation](#).
- [3] Shuhong Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222v1 (2026), especially Theorems 3.3–3.4 and Appendix A.1. <https://arxiv.org/html/2608.00222v1>.
- [4] royvanrijn/jacobian-research, *Exact finite-field value distribution*, archived research note, research/archive/legacy-notes/FINITE_FIELD_VALUE_DISTRIBUTION.md. Repository snapshot caf5685db1099cb90e608556d4b8d825734f29cb, inspected September 24, 2026. [Pinned finite-field note](#). The repository note is cited as prior public work, not as an independently peer-reviewed source.
- [5] royvanrijn/jacobian-research, *An integral Jacobian-one Hasse failure*, archived research note, research/archive/non-elliptic/verified/INTEGRAL_HASSE_JACOBIAN_ONE.md, same pinned snapshot. [Pinned degree-five note](#). Its distinct map and rational obstruction are used only for comparison, not as assumptions in any theorem of this article.
- [6] James S. Milne, *Algebraic Number Theory*, version 3.08, July 19, 2020, Theorem 8.31 and Example 8.36. <https://www.jmilne.org/math/CourseNotes/ANT.pdf>.